

Permutation and Combination

1. A Committee of 5 members is to be formed from 10 people. If A and B must always be together, how many different committees can be formed?

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A and B must always be together, they are fixed in the committee.

- Total people = 10
- A and B are already selected. = 2
- Remaining people = $10 - 2 = 8$
- Remaining member to be selected = $5 - 2 = 3$

Using the Combination formula:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^8C_3 = \frac{8!}{3!(8-3)!} = \frac{8!}{3! \times 5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = \frac{336}{6} = 56$$

∴ 56 Committees.

2. From 8 students, a team of 4 is to be selected. If A must be included, how many teams can be formed?

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A must be included, A is fixed in the team.

- Total students = 8
- A is already selected
- Remaining students to choose = $4 - 1 = 3$
- Remaining students = $8 - 1 = 7$

Using the combination formula:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^7C_3 = \frac{7!}{3!(7-3)!}$$

$$= \frac{7!}{3! \times 4!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

$$\therefore 35 \text{ teams.} = \frac{210}{6}$$

3. From 12 Players, a team of 5 is to be selected. If a Particular Player A cannot be selected, how many teams can be formed?

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A cannot be selected, exclude A from the selection.

• Total Players = 12

• Excluding A leaves = 11 Players.

• Select 5 Players from these 11.

Using the combination formula:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^{11}C_5 = \frac{11!}{5!(11-5)!}$$

$$= \frac{11!}{5! \times 6!} = \frac{11 \times 10 \times 9 \times 8 \times 7}{5 \times 4 \times 3 \times 2 \times 1} = 462$$

$$= \frac{55440}{120}$$

$\therefore 462$ teams.